UPGRADING OF EXISTING DCS SYSTEMS & ASSOCIATED EQUIPMENT FOR CYBER SECURITY COMPLIANCE AT RMP CCR & OMCCR IN MAA REFINERY

CYBERSECURITY - OVERALL ZONES AND CONDUITS DRAWING (RMP)

REV	DATE (DD.MM.YY)	APPRD BY	DESCRIPTION	PROJECT ORDER NUMBER	PROJ.ENG.	SPEC.ENG.
C	29.05.2025	M.A	RE-ISSUED FOR APPROVAL	CA/CPC/0707		
B	20.04.2025	M.A	ISSUED FOR APPROVAL	CA/CPC/0707		
A	10.12.2024	M.A	ISSUED FOR REVIEW	CA/CPC/0707		

KNPC APPROVAL

MASTER DRAWING REVISION

AREA: RMP CCR & OMCCR IN MAA REFINERY

SUB AREA: RMP — NO:

DRWG.TYPE: ZONES AND CONDUITS DRAWING

DESCRIPTION: OVERALL ZONES AND CONDUITS DRAWING

SCALE : NTS

PROJECT ORDER NUMBER : CA/CPC/0707

DRAWING: ME-20420-RMP-ZCD-001

شركة البترول الوطنية الكويتية (ش.م.ك)
KNPC

Schneider Electric

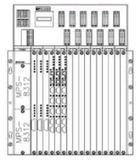
ASSET CODE	UNIT CODE	ORIGINATOR CODE	DISC. CODE	DOC. TYPE.	SEQUENCE NO.	SHEET NO.	REV. NO.

ABBREVIATIONS:

ABBREVIATION	DESCRIPTION
AIMAPI	AIM Application Interface
ALMS	Alarm management system
AMS	Asset management system
APC	Advanced process control
CA	Certificate authority
DMZ	Demilitarized zone
DWS	Display wal system
ePO	ePolicy Orchestrator (Trellix endpoint protection)
ESD	Emergency Shutdown System
FCP	Field Control Processor (FCP270 or 280)
FDC	Field Device Controller (FDC280)
FDCN	Foxboro DCS Control Network
FGS	Fire and Gas System
GW	Gateway
IDS	Intrusion detection system
LAN	Local area network
NPM	Network performance monitoring
OOB	Out of band
OPC	OLE for process control
P2P	Peat to peer
PDC	Primary domain controller
PIN	Plant information network
RADIUS	Remote Authentication Dial-in User Service
SA	System Advisor
SDC	Secondary domain controller
SIS	Safety Instrumented System
TC	Thin client
TCP/IP	TRANSMISSION CONTROL PROTOCOL/INTERNET PROTOCOL
TSAA	Triconex System Access Application
VHN	Virtualization host network
VM	Virtual machine
WSUS	Windows server update services

Legend	Description
— · — · — · — · — · — · —	FIBRE OPTIC CABLE (FO)
- - - - - - - - - - - - - - -	ETHERNET CABLE
	LEVEL 1 (PROCESS CONTROL LEVEL)
	LEVEL 2.0 (FOXBORO EVO CONTROL NETWORK)
	LEVEL 3.0 - PLANT INFORMATION NETWORK
	LEVEL 3.2 - SECURITY NETWORK
	LEVEL 3.5 - DMZ
	LEVEL 4
	VHN NETWORK
	SPAN TRAFFIC

LEGENDS:



TRICONEX/TRIDENT CONTROLLERS

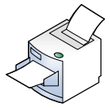
VIRTUALIZATION HOST / SERVER

FIREWALL

NOZOMI IDS

THIN CLIENT

PHYSICAL MACHINE



PRINTER

NTP SERVER

AGGR

AGGREGATOR SWITCH (IDS)

UNTRUSTED BOUNDARY

CONTROL SYSTEM ZONE

THIRD PARTY ZONE

CYBERSECURITY ZONE

SCHNEIDER SIS OR GDS ZONE

CONDUIT

ACCESS LIST

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KNPC APPROVAL

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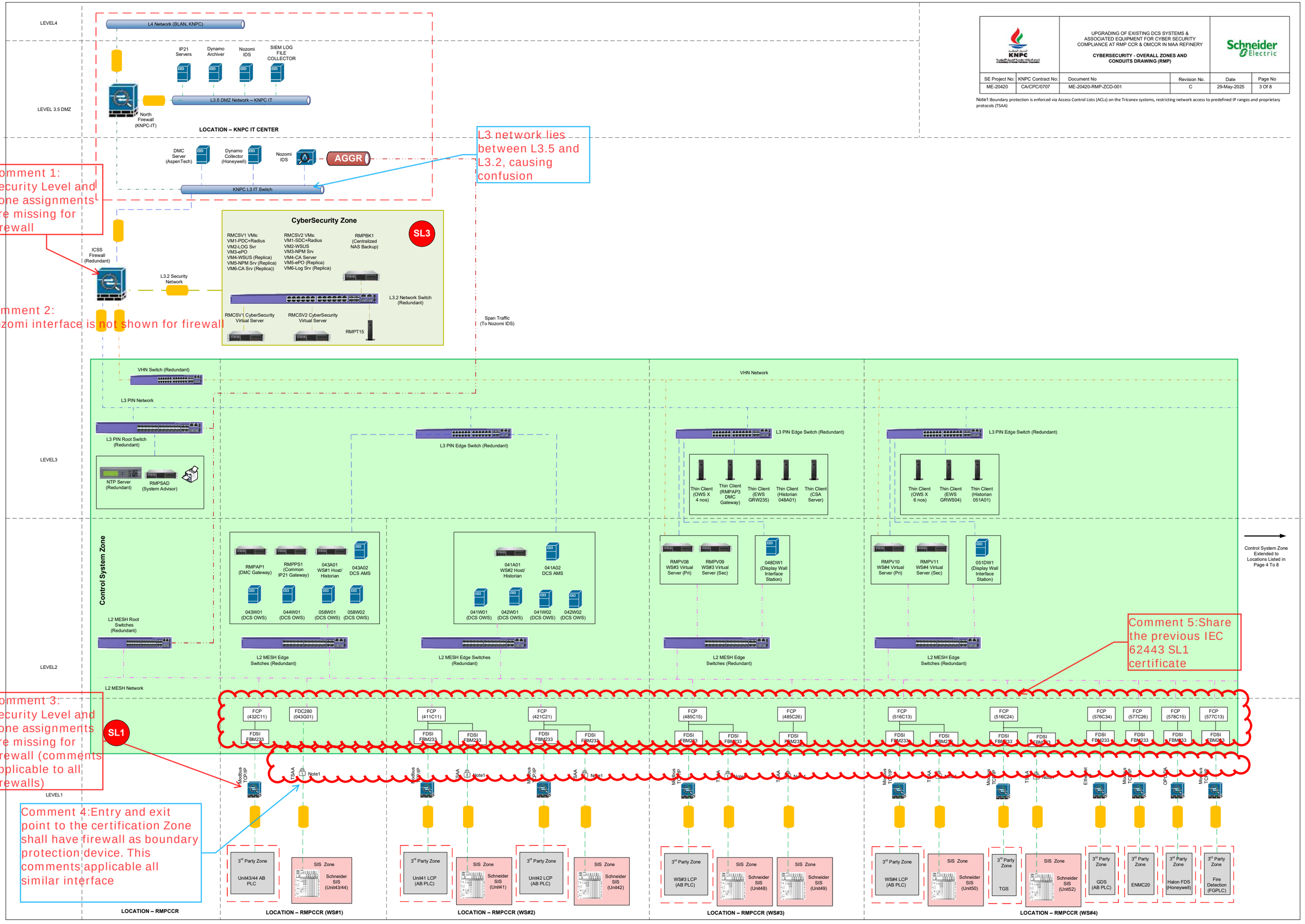
SE Project No:	KNPC Contract No:	Document No	Revision No.	Date	Page No
ME-20420	CA/CPC/0707	ME-20420-RMP-ZCD-001	C	29-May-2025	3 Of 8

Note1:Boundary protection is enforced via Access Control Lists (ACLs) on the Triconex systems, restricting network access to predefined IP ranges and proprietary protocols (TSAA)



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SE Project No:	KNPC Contract No:	Document No	Revision No.	Date	Page No
ME-20420	CA/CPC/0707	ME-20420-RMP-ZCD-001	C	29-May-2025	4 Of 8

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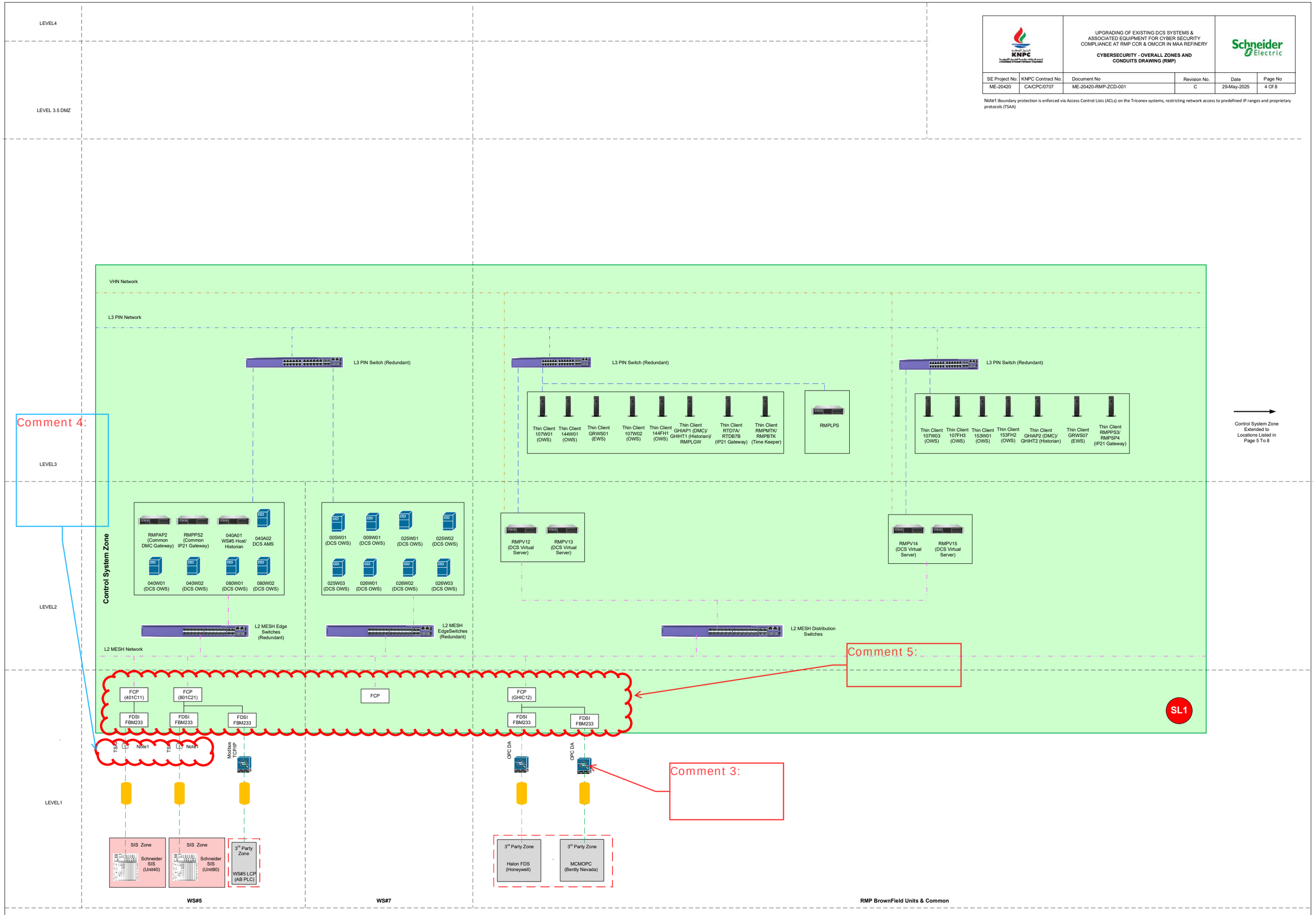

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SE Project No:	KNPC Contract No:	Document No	Revision No.	Date	Page No
ME-20420	CA/CPC/0707	ME-20420-RMP-ZCD-001	C	29-May-2025	5 Of 8

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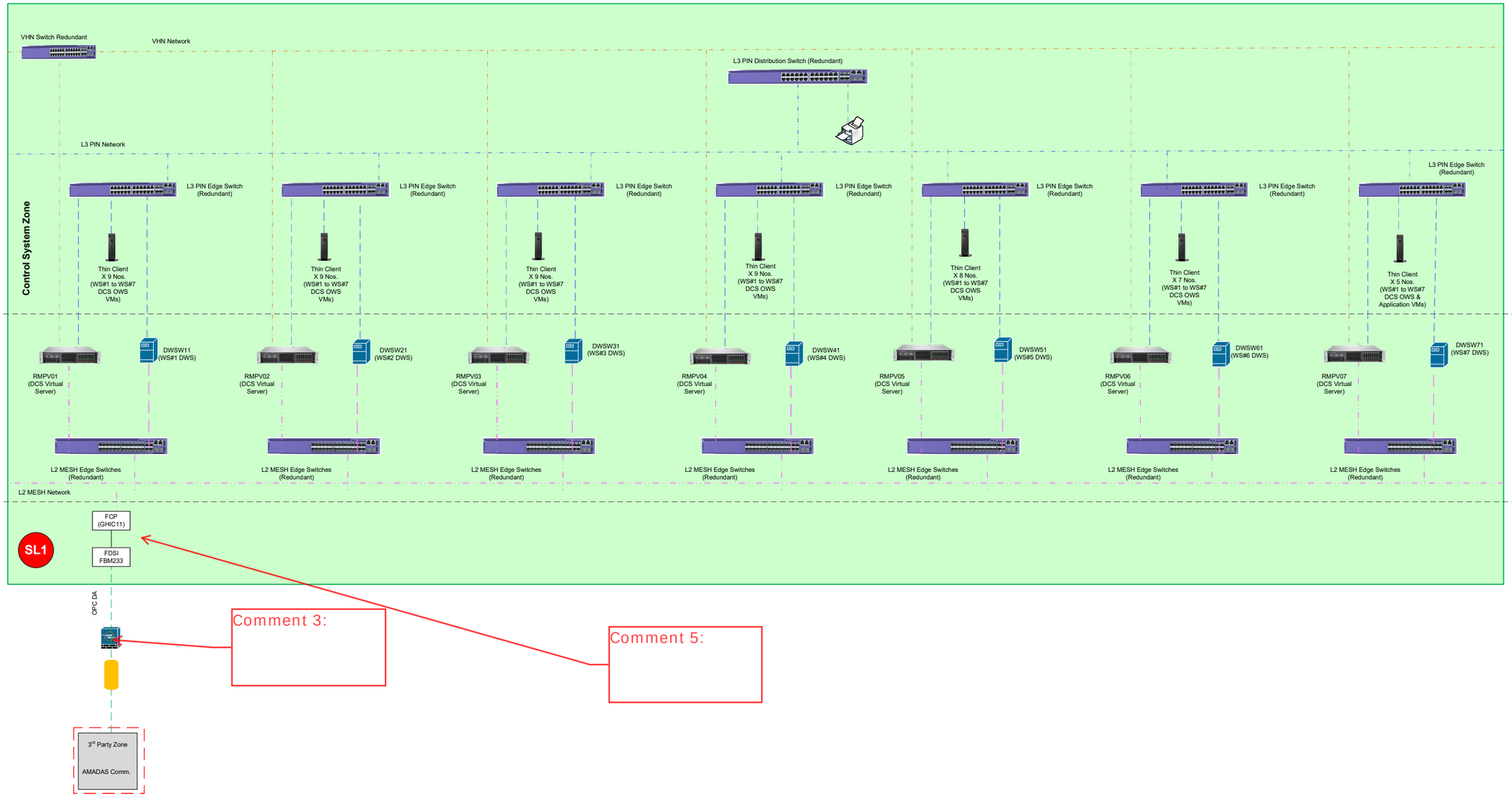

LOCATION – RCC (WS#1 to WS#7)

KNPC		UPGRADING OF EXISTING DCS SYSTEMS & ASSOCIATED EQUIPMENT FOR CYBER SECURITY COMPLIANCE AT RMP CCR & OMCCR IN MAA REFINERY **CYBERSECURITY - OVERALL ZONES AND CONDUITS DRAWING (RMP)**		Schneider Electric	
SE Project No:	KNPC Contract No:	Document No	Revision No.	Date	Page No
ME-20420	CA/CPC/0707	ME-20420-RMP-ZCD-001	C	29-May-2025	6 Of 8

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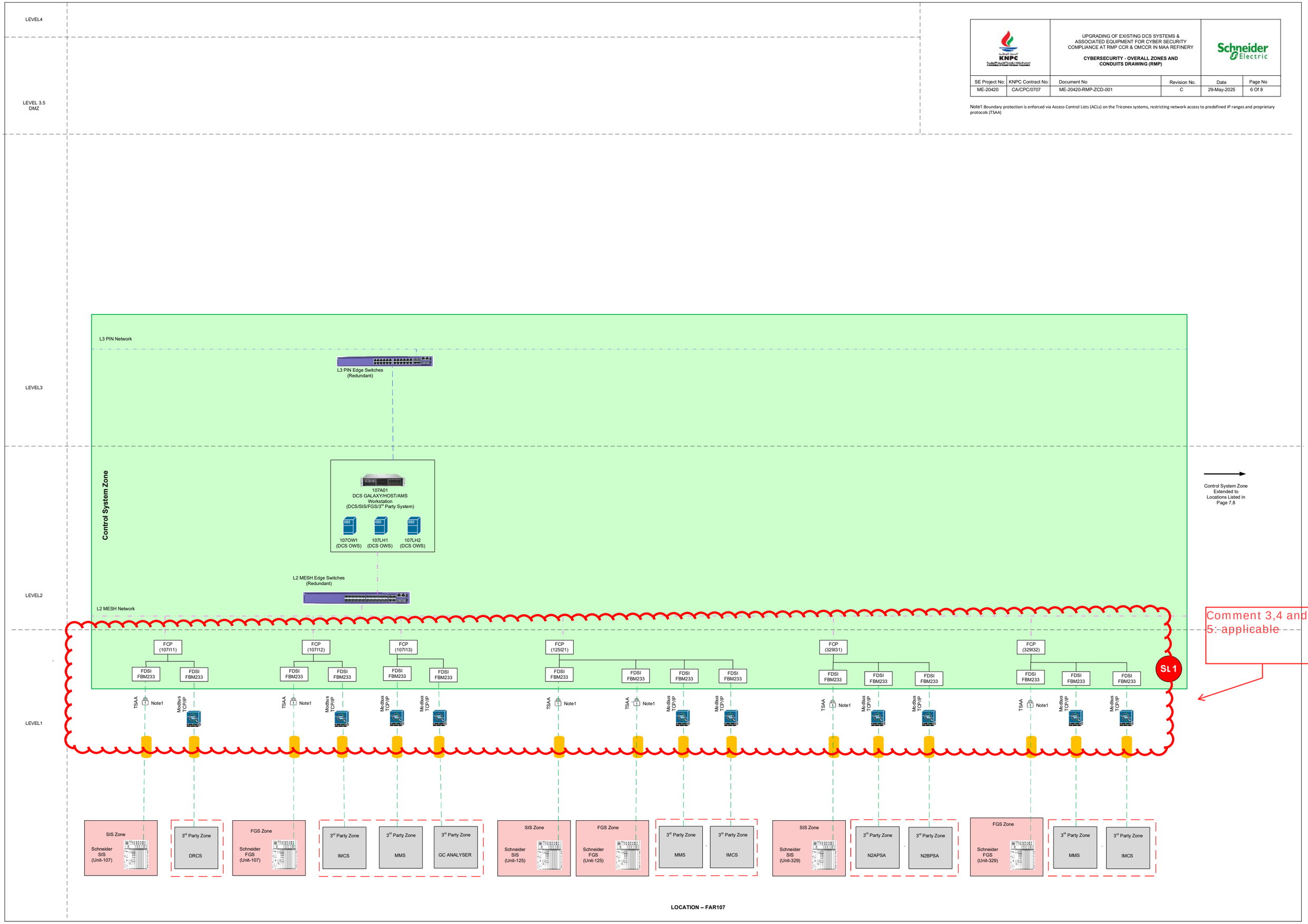

LOCATION – FAR107

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SE Project No:	KNPC Contract No:	Document No	Revision No.	Date	Page No
ME-20420	CA/CPC/0707	ME-20420-RMP-ZCD-001	C	29-May-2025	7 Of 8

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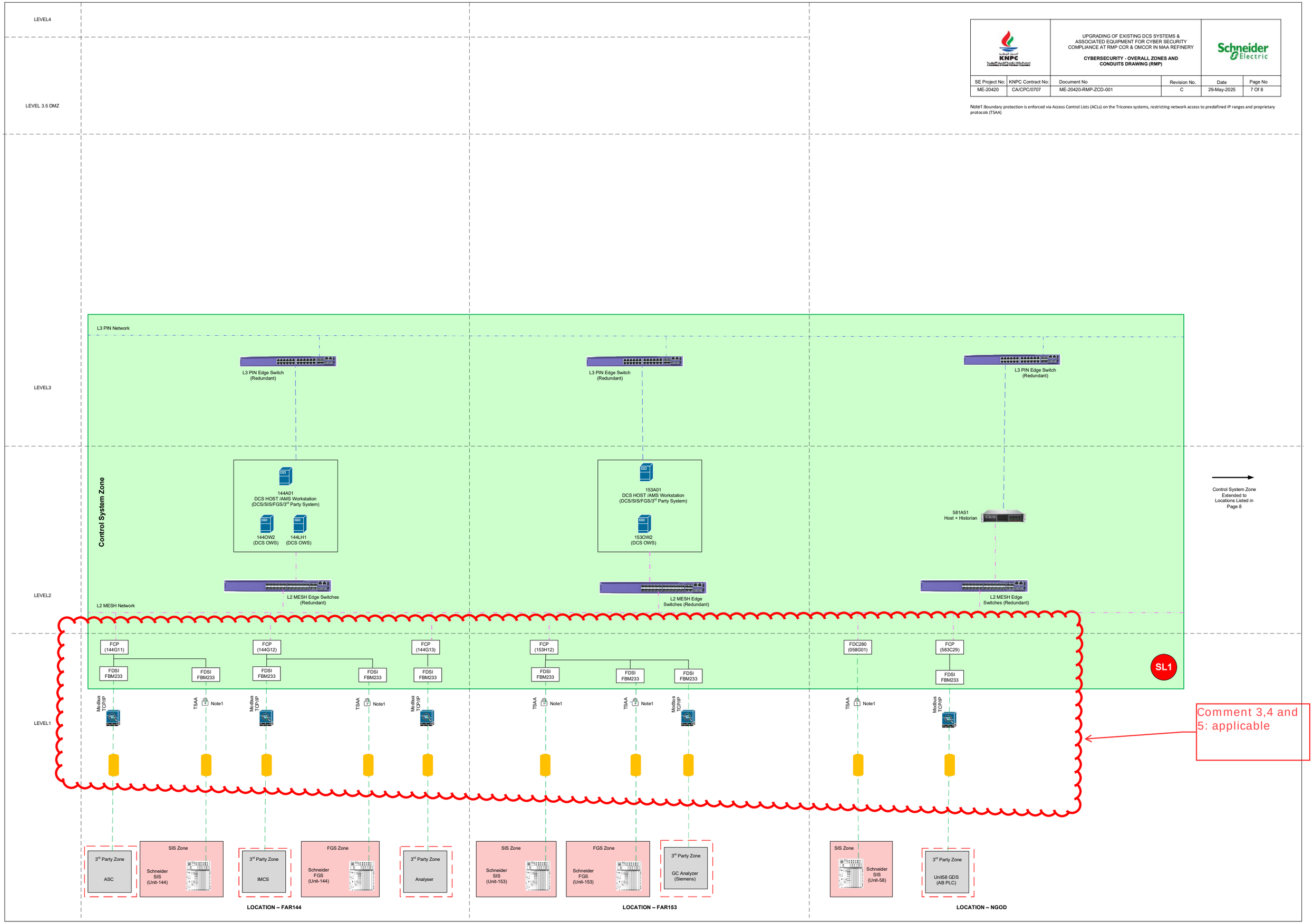

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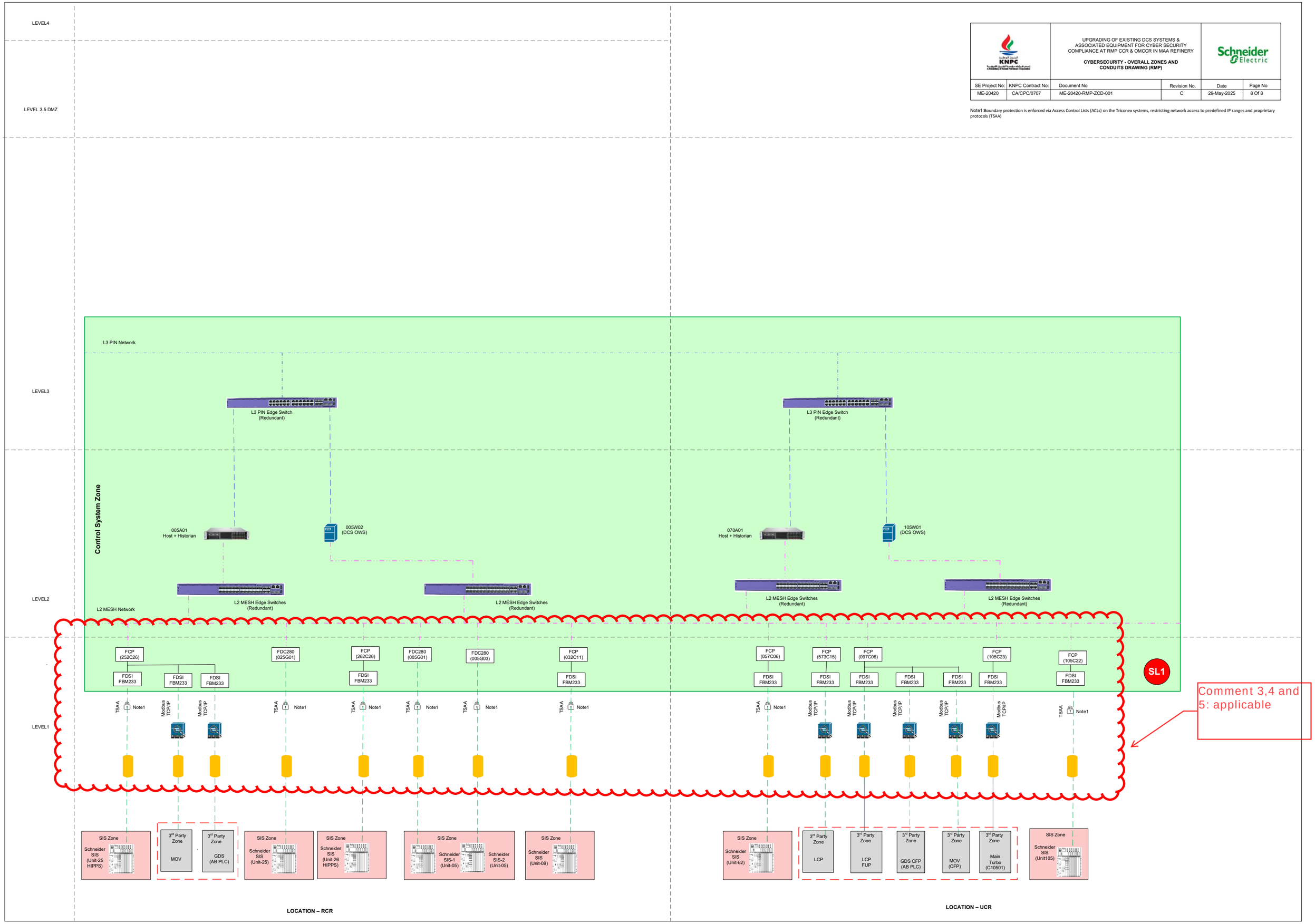
ME-20420	CA/CPC/0707	ME-20420-RMP-ZCD-001	C	29-May-2025	8 Of 8

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